



Fungal evasion of *Drosophila* immunity involves blocking the cathepsin-mediated cleavage maturation of the danger-sensing protease

Guirong Tang^{a,b} , Shuangxiu Song^a, Junmei Shang^a , Yujuan Luo^{a,c} , Shiqin Li^{a,c}, Dongxiang Wei^{a,d} , and Chengshu Wang^{a,c,d,1}

Affiliations are included on p. 9.

Edited by Joseph Heitman, Duke University School of Medicine, Durham, NC; received September 20, 2024; accepted December 23, 2024

Entomopathogenic fungi play a critical role in regulating insect populations, and representative species from the *Metarhizium* and *Beauveria* genera have been developed as eco-friendly biocontrol agents for managing agricultural insect pests. Relative to the advances in understanding antifungal immune responses in *Drosophila*, knowledge of how fungi evade insect immune defenses remains limited. In this study, we report the identification and characterization of a virulence-required effector Fkp1 in *Metarhizium robertsii*. Library screening and protein pull-down analysis unveiled that Fkp1 targets the cathepsin protease CtsK1 to inhibit its cleavage maturation of the danger-sensing serine protease Persephone (Psh), thereby facilitating fungal evasion of the *Drosophila* immune defenses. The *Fkp1*-like gene is also required in *Beauveria bassiana* for insect infection. Transgenic expression of *Fkp1* in *Drosophila* suppressed hemolymph cysteine protease activity and down-regulated the expression of antifungal genes. Fkp1 can also mask the Psh cleavage site without interfering with its ability to bait fungal subtilisin proteases. Given the evident compensatory relationship, our data indicate that the protease cascade is more crucial than the molecular pattern pathway in defending flies against fungal infections. This work reveals that *Metarhizium* fungi have evolved distinct effectors to block the dual recognition pathways of flies for immune evasion and sheds lights on the effector mechanisms mediating microbe–animal interactions.

Metarhizium | effector | antifungal immunity | cathepsin | Persephone

Entomopathogenic fungi (EPF) act as one of the key factors in regulating insect populations in nature. Certain species such as the *Metarhizium robertsii* and *Beauveria bassiana* have been utilized as genetically tractable models for investigating fungus–insect interactions and have been developed into eco-friendly insect biocontrol agents (1, 2). Compared to the advancements in understanding effector-mediated fungus–plant interactions (3), the strategies employed by EPF to subvert insect immune defenses remain relatively unexplored (1). In support of the emerging evidence of gene-for-gene relationships between microbes and metazoans (4, 5), effector-mediated immune evasions have been clarified in fungus–insect interactions. For example, similar to those chitin-binding lysin motif (LysM) containing effectors encoded in plant pathogens (6), the LysM proteins of the insect pathogen *B. bassiana* can similarly bind to chitin polymers or oligomers for camouflaging the cell wall pathogen-associated molecular patterns (PAMPs) within the insect body cavity (7). We have recently shown that the Tge1 effector encoded by *M. robertsii* can block the β -glucan coreceptors GGBP3 and GGBP-like 3 (GL3) of *Drosophila melanogaster* to facilitate immune evasion (8). *Drosophila* has also evolved the serine protease Persephone (Psh) to sense the danger signal of pathogen proteases (9–11), which is matured by cleavage of the cysteine protease (abbreviated CysP) 26–29-p (now called CtsK1) to activate the antifungal Toll pathway (11). It is unclear whether fungal parasites can disrupt this pathway during infection processes.

Different families of CysPs are omnipresent and play essential physiological roles in different organisms (12, 13). In plants, papain-like CysPs function as hubs in immunity (14). Thus, different plant pathogenic fungi have evolved distinct effectors to block host immune CysPs. For example, the avirulence factor Avr2 of the pathogenic fungus *Cladosporium fulvum* inhibits the tomato Rcr3, a CysP conferring the induction of hypersensitive response (15). The Pit2 effector of the corn smut fungus *Ustilago maydis* has been identified as an inhibitor of the papain-like CysPs linked with the salicylic-acid-mediated plant defenses (16, 17). A species-specific effector MoErs1 from the rice blast fungus *Magnaporthe oryzae* has been characterized for its ability to inhibit the rice CysP OsRD21, a key player in rice immune defenses (18). Likewise, the effector

Significance

The dual sensing pathways mediating antifungal immunity in *Drosophila* have been clarified: circulating receptors that recognize fungal cell wall components and a protease cascade that detects fungal danger signals. While these pathways are effective, the strategies employed by fungi to evade host antifungal immune surveillance remain elusive. This study reveals that similar to plant pathogens, *Metarhizium* fungi utilize a gene-for-gene mechanism to suppress insect antifungal immunity. Specifically, *Metarhizium robertsii* secretes an effector that targets *Drosophila* hemolymph cysteine protease, thereby blocking the danger-sensing cascade. Combined with the previous finding of effector-mediated blocking of β -glucan receptors, our findings reveal that *Metarhizium* fungi have evolved distinct effectors to target insect immune sensors.

Author contributions: C.W. designed research; G.T., S.S., J.S., Y.L., S.L., and D.W. performed research; G.T. and C.W. analyzed data; and C.W. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

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¹To whom correspondence may be addressed. Email: wangcs@sippe.ac.cn.

This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2419343122/-/DCSupplemental>.

Published January 16, 2025.

AVRblb2 secreted by the oomycete pathogen *Phytophthora infestans* targets the host CysP C14 to prevent its entry into apoplast for immune defenses (19). In addition to the *Drosophila* CtsK1 (11), animal CysPs of the papain family (named cathepsins) also have diverse biological functions including immune response (20, 21). A cystatin-like CysP inhibiting protein encoded by a symbiotic virus of the parasitoid *Cotesia congregata* can promote wasp parasitic success by targeting the 26–29-p CysP of the caterpillar *Manduca sexta* (22). Until this study, no inhibitor has been identified from EPF in targeting insect cathepsins.

During the screening of virulence-related effector genes in *M. robertsii*, a putative surface antigen gene *EXU99903* was identified as essential for fungal infection of caterpillars for an unclear mechanism (8). In this study, we report that this gene (termed *Fkp1* for a fungus effector kidnapping protease cascade) contributes to blocking the cathepsin CtsK1 from cleavage maturation of Psh in *Drosophila*, thereby suppressing fly antifungal immune responses.

Results

Fkp1 Is Required for Fungal Virulence to Suppress *Drosophila* Immune Responses. We first performed the Blast search of *Fkp1*-like genes encoded by different fungi and found that its homologue is patchily distributed in the Pezizomycotina subphylum of Ascomycota, including its presence in different *Metarhizium* species and the other insect, plant, and nematode pathogenic fungi (Fig. 1A). For example, *B. bassiana* has an *Fkp1* homologue (BBA_10113, 37% identity at the amino acid level; termed BbFkp1). Intriguingly, the *Fkp1* homologue is absent in the *Beauveria*-closely related EPF such as *B. brongniartii*, *Cordyceps militaris*, *C. cicadae*, and *Ophiocordyceps sinensis* that are mild/slow in infecting and killing certain insect species (23, 24). It is noteworthy that the *Fkp1*-like genes are also absent in mammalian pathogenic fungi such as *Aspergillus fumigatus* and *Candida* species.

After growing the fungus at different conditions, our reverse-transcription quantitative PCR (RT-qPCR) analysis revealed that *Fkp1* was highly transcribed by *M. robertsii* during

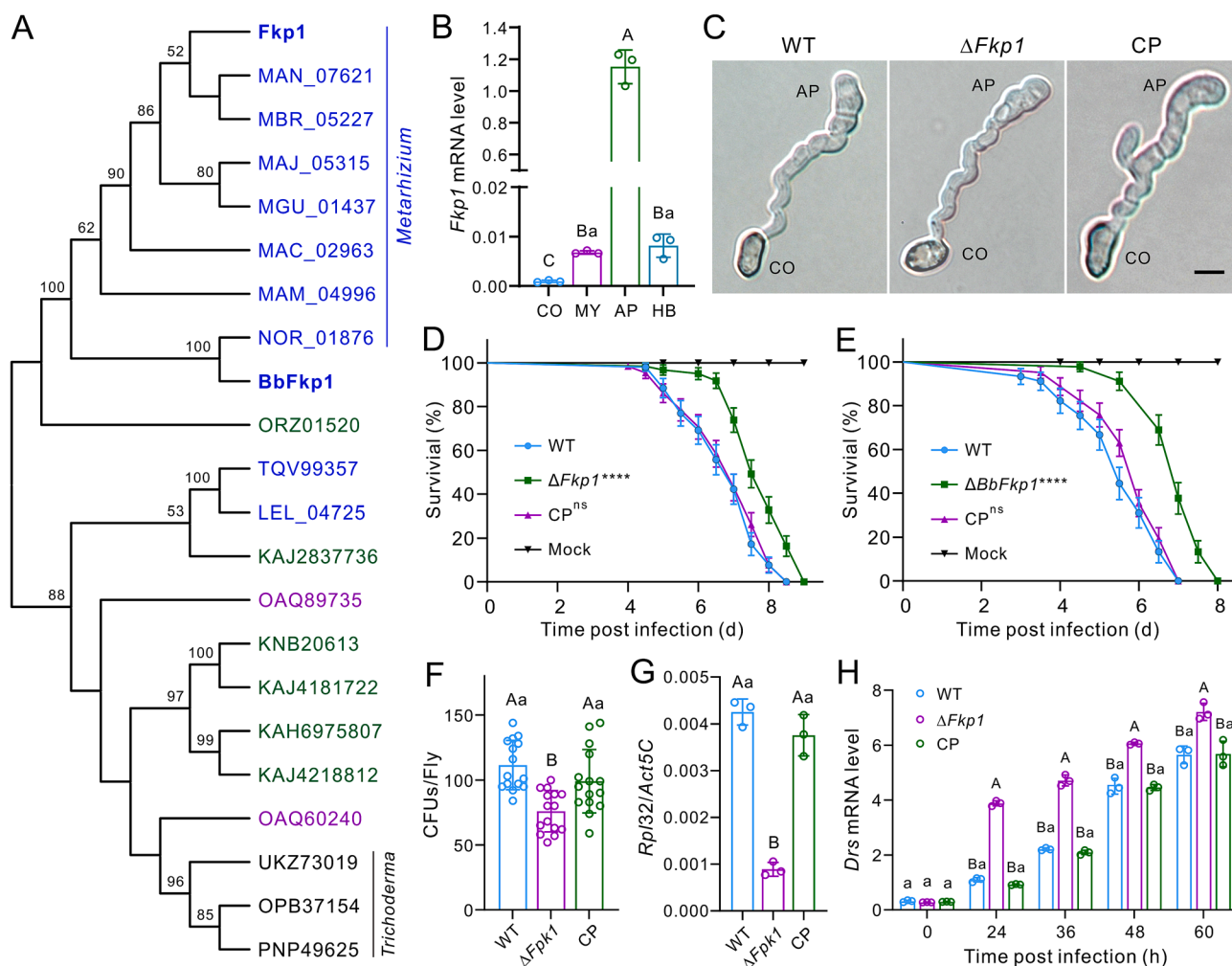


Fig. 1. The conserved effector gene is required for fungal virulence by suppressing fly antifungal immunity. (A) Phylogeny analysis of the *Fkp1*-like genes from different fungi. The consensus maximum likelihood tree was generated based on the aligned protein sequences. Accessions labeled in blue are from EPF; those in green from plant pathogenic fungi; those in pink from nematophagous fungi and in black from the mycoparasitic fungi. (B) Differential expression of *Fkp1* by *M. robertsii* at different stages. CO, conidium; MY, mycelia harvested from SDB; AP, appressoria formed on fly wings for 30 h; HB, hyphal bodies harvested from the *Galleria* larvae hemolymph. (C) Deletion of *Fkp1* in *M. robertsii* did not affect the formation of appressoria on a hydrophobic surface. CP, complemented strain. (Scale bar, 5 μ m.) (D and E) Deletion of *Fkp1* in *M. robertsii* (D) and *BbFkp1* in *B. bassiana* (E) significantly impaired fungal virulence against *Drosophila* females. Values are plotted as the mean $\pm$ SEM. The difference in fly survivals was compared between the WT and individual mutants by the log-rank test: * P < 0.05, ** P < 0.01, **** P < 0.0001. ns, not significant. (F and G) Fungal load assays determined by CFU counting (F) and qPCR analysis of fungal DNA copies (G) 84 h after topical challenge of female flies with the WT *M. robertsii* spores. (H) Deletion of *Fkp1* impaired fungal ability to suppress the expression of *Drs* in *D. melanogaster*. Panels B and F–H: Values are the mean $\pm$ SD. Different capital (P < 0.01) or lower (P < 0.05) letters labeled above columns show the results of the one-way ANOVA followed by Tukey's test. Note for panel H, Differences were compared between samples examined at the same time point. Unless otherwise indicated, the same statistical analysis was conducted for the subsequent experiments.

the formation of the infection structure appressoria on fly wings, which was followed by gene expression in fungal cells growing in the body cavities of wax moth larvae (*Galleria mellonella*) (Fig. 1B). By employing a yeast secretion-trap method (25), we verified that the signal-peptide containing Fkp1 has a secretion feature (SI Appendix, Fig. S1A).

By using the previously obtained *Fkp1* null mutant as a recipient (8), we generated the gene-complemented strain for further experiments (SI Appendix, Fig. S1B). In addition, the *BbFkp1* gene was deleted and the null mutant was rescued in *B. bassiana* (SI Appendix, Fig. S1C). Deletion of *Fkp1* did not affect the formation of appressoria (Fig. 1C), saprophytic growth, and penetration of the cellophane membrane (SI Appendix, Fig. S1D). However, it was confirmed that *Fkp1* is essentially required for *Metarhizium* infection of *Drosophila* females (log-rank test, $P < 0.0001$) (Fig. 1D) and males (SI Appendix, Fig. S1E). Likewise, the null mutant of *BbFkp1* was significantly impaired ($P < 0.0001$) in infecting female and male flies (Fig. 1E and SI Appendix, Fig. S1F). Similar results were obtained in infection of the spotted wing drosophila *D. suzukii* females and males using wild-type (WT) and mutant strains of *M. robertsii* and *B. bassiana* (SI Appendix, Fig. S1H–K). We also performed an overexpression of *Fkp1* in *M. robertsii* (SI Appendix, Fig. S1L), and subsequent survival assays revealed that the transgenic mutants could more quickly ($P < 0.0001$) kill flies than the WT did (SI Appendix, Fig. S1M). After confirming the virulence requirement of *Fkp1/BbFkp1* in two fungi, subsequent experiments were conducted using only *M. robertsii* strains unless stated otherwise.

In support of virulence defect, our fungal load assays revealed that $\Delta Fkp1$ had a significantly reduced ability (one-way ANOVA, $P < 0.01$) to colonize flies when compared with the WT and gene-rescued strains by counting fungal colony-forming units (CFUs) and qPCR analysis of fungal DNA copies in flies (Fig. 1F and G). After the topical challenge of flies for different durations of time, our analysis of the antifungal *Drosomycin* (*Drs*) gene revealed that the deletion of *Fkp1* substantially reduced fungal ability ($P < 0.01$) to suppress fly immune defenses (Fig. 1H).

Fkp1 Targets the Cathepsin CtsK1 for Suppressing Cysteine Protease Activity. To identify the potential target of Fkp1, we cloned its cDNA as bait to screen the *D. melanogaster* cDNA library (26). The positive clones were selected for PCR sequencing of inserted genes. As a result, nine fly genes harbored in twenty yeast clones were obtained (SI Appendix, Table S1). The subsequent yeast-two hybrid (Y2H) analysis verified that Fkp1 could positively interact with two cathepsin proteases, i.e., CtsK1 and CtsB, of *D. melanogaster* (Fig. 2A). The *Drosophila* genome encodes ten modular cathepsins (SI Appendix, Fig. S2A and Table S2). For example, the circulating CtsK1 contains the N-terminal cathepsin propeptide inhibitor domain I29 and the C-terminal catalytic C1A peptidase domain (Fig. 2B). Further Y2H analysis revealed that Fkp1 only targeted the C terminus (CtsK1C) but not the N-terminal region (CtsK1N) of CtsK1, and it was also true for BbFkp1 (Fig. 2C). Y2H interactive analysis with the other cathepsins showed that both Fkp1 and BbFkp1 could potentially target the conserved peptidase domains of CtsL1, CtsL3, CstK2, CtsF, and Swim of *D. melanogaster* (SI Appendix, Fig. S2B–D).

We next performed the heterologous expression of the tag-fused Fkp1 and CtsK1C (SI Appendix, Fig. S3), and subsequent protein pull-down analysis confirmed the interaction between Fkp1 and CtsK1C (Fig. 2D). After the topical challenge of flies with different *Metarhizium* strains, insect hemolymph samples were collected for CysP activity assays. The results revealed that total cathepsin activity was significantly higher ($P < 0.01$) in flies infected with

$\Delta Fkp1$ compared to those infected with the WT and complemented strains at different time points (Fig. 2E), confirming that Fkp1 contributes to suppressing cathepsin activities during fungal colonization of insects.

CtsK1 Is Required in *Drosophila* for Mediating Antifungal Immunity. First, our survival assays did not find the obvious difference between *CtsB* Δ and W1118 flies after challenge with the WT strain of *M. robertsii* (SI Appendix, Fig. S2E). The *CtsB* gene was then not included in further analysis. We next had the *CtsK1* gene disrupted using a CRISPA-Cas9 technique. Two independent mutants were obtained and verified to generate the isogenic mutant lines after self-crossing for more than five generations (Fig. 3A). Enzyme activity assays revealed that the disruption of *CtsK1* substantially reduced ($P < 0.01$) the fly hemolymph cathepsin activity before and after topical challenge with the WT *M. robertsii* (Fig. 3B), indicating that CtsK1 represents a major circulating CysP in flies. By including the *Psh*¹ (27), *GNBP3* Δ , and *GL3* Δ (8) mutant flies, our topical infections of both female and male flies revealed that *CtsK1* is crucially required ($P < 0.0001$) in flies against the topical infection of *M. robertsii* based on the survivals of two independent mutant fly lines (Fig. 3C and D). Similar results were obtained after topical infection of female flies with *B. bassiana* (SI Appendix, Fig. S4A). In addition, it was evident that *CtsK1* is more essential ($p < 0.0001$) than *GNBP3* but less important than *Psh* ($P < 0.0001$) in *Drosophila* in defending flies against *Metarhizium* infection (Fig. 3C and D and SI Appendix, Table S3). Relative to the fungal loads detected in the W1118 flies, both *CtsK1* null mutants were impaired ($P < 0.0001$) in curbing *Metarhizium* colonization (Fig. 3E). Consistently, expression of the antifungal genes *Drs* and *Baramycin A* (*BaraA*) (28) was substantially reduced ($P < 0.01$) in the *CtsK1* mutant flies after topical fungal infection for different durations (Fig. 3F and G).

To determine the *CtsK1* disruption effect on the fly control of binary sensing pathways, we examined the accumulation of GNBP3, GL3, and the cytokine Spätzle (Spz) processing protease Spe in fly hemolymph (SI Appendix, Fig. S5A). It was found that relative to those in the W1118 flies, the expression of GNBP3 was highly activated in both *CtsK1* mutant flies, which was even stronger than that in the *Psh*¹ flies (Fig. 3H). For example, the expression of this surveillance receptor was up-regulated by more than 6-fold in *CtsK1* Δ mutants versus its 4.12-fold upregulation in *Psh*¹. However, the expression of the inductive β -glucan coreceptor GL3 was not obviously impaired in the *CtsK1* Δ mutant flies while GL3 demonstrated a delayed induction in *Psh*¹ (Fig. 3I). Consistently, the observations were largely corroborated by our RT-qPCR analysis of *GNBP3* and *GL3* transcriptions in different flies (SI Appendix, Fig. S4B and C). The detection of the downstream Spe showed that this Spz ligand-processing protease was down-regulated in the *CtsK1* Δ mutant flies and almost abolished in *Psh*¹ (Fig. 3J).

Transgenic Expression of *Fkp1* in *Drosophila* Counters Fly Antifungal Immunity. Next, we performed the transgenic insertion of *Fkp1* into either the chromosome II [UAS-*Fkp1*(II)] or III [UAS-*Fkp1*(III)] of *D. melanogaster*. The subsequent topical infections with both *M. robertsii* and *B. bassiana* revealed that both the transgenic females succumbed much faster ($P < 0.001$) than the mock controls upon fungal challenges (Fig. 4A and B). In support, fungal load assays confirmed that the transgenic flies were significantly impaired ($P < 0.0001$) in their abilities to counter fungal colonization (Fig. 4C). CysP activity assays showed that upon topical fungal infection, the transgenic expression of *Fkp1* in flies could considerably down-regulate

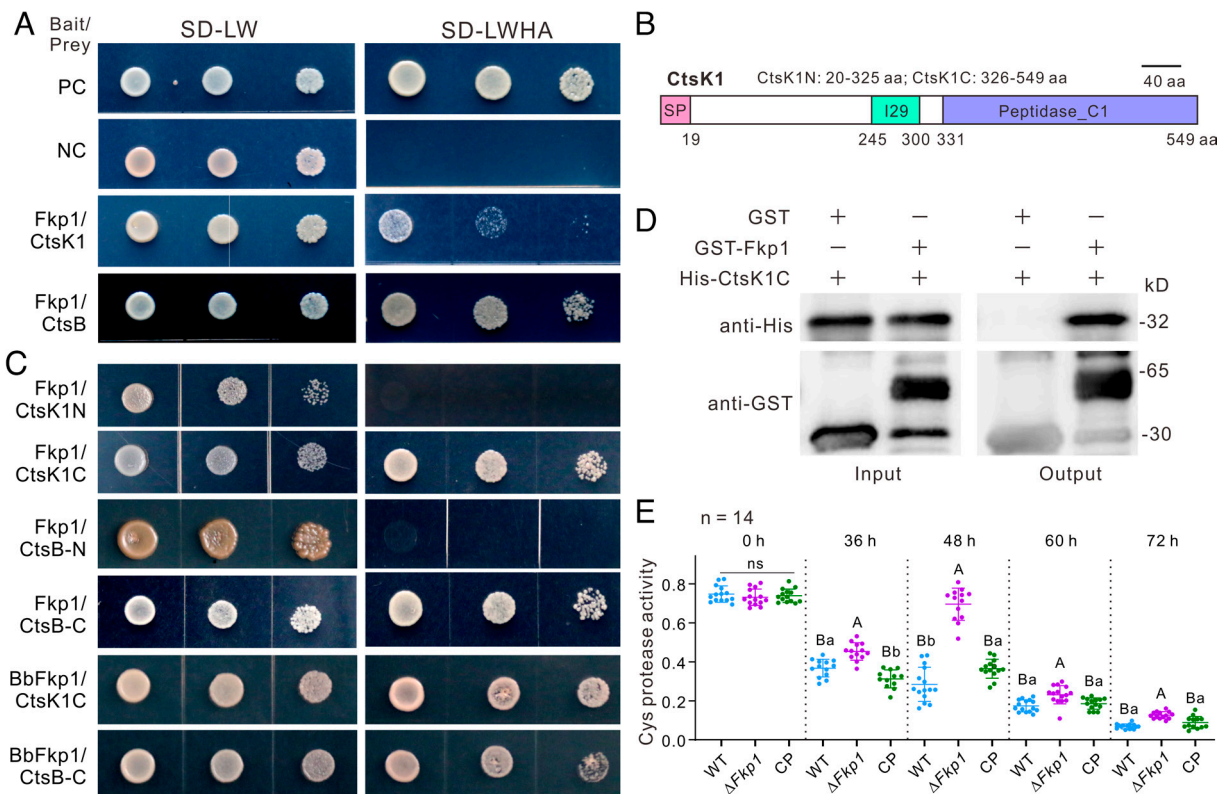


Fig. 2. Confirmation of Fkp1 in targeting the cysteine proteases of *D. melanogaster*. (A) Y2H analysis confirms the interaction between Fkp1 and CtsK1/CtsB. (B) Schematic structuring of the *Drosophila* CtsK1. SP, signal peptide; I29: Cathepsin propeptide inhibitor domain. The N- and C-terminal regions used for the following experiments are indicated. (C) Y2H analysis shows that fungal effectors only target the peptidase domain of CtsK1 and CtsB. (D) Protein pull-down analysis confirms the interaction between Fkp1 and the peptidase domain of CtsK1. (E) Deletion of *Fkp1* impaired fungal ability to suppress the circulating cathepsin protease activity in *D. melanogaster*. After topical fungal infections, fly hemolymph was collected for activity assays using a fluorescent substrate. Enzyme unit activity was determined as the release of unit fluorescence per min per microgram of protein.

($P < 0.01$) the cathepsin activity when compared with the mock controls (Fig. 4D). RT-qPCR analysis unveiled that the expression of *Drs* and *BaraA* was down-regulated ($P < 0.01$) in the transgenic flies when topically infected by *M. robertsii* (Fig. 4E and F). Overall, no obvious differences were observed between the two transgenic fly lines.

We also detected the differential accumulation of the above immune proteins in the *Fkp1*-transgenic fly hemolymphs after topical fungal infection for different durations (SI Appendix, Fig. S5B). It was found that the transgenic expression of *Fkp1* resulted in the considerable upregulation of GGBP3 in flies compared to mock controls (Fig. 4G and SI Appendix, Fig. S4D). Rather like its expression patterns in the *CtsK1* mutants, GL3 induction had no obvious difference between the mock and *Fkp1*-transgenic flies (Fig. 4H). Similar to the *CtsK1* null mutants, both the *Ubi>Fkp1*(II) and *Ubi>Fkp1*(III) flies had reduced accumulation of Spe after fungal infections (Fig. 4I).

Fkp1 Blocks the CtsK1-Mediated Cleavage Activation of Psh. As reported before (11, 29), CtsK1 mediates the maturation of Psh by cleavage at the VIH/IVGGY site of its N terminus (PshN). PshN contains a signal peptide and a CLIP domain (termed PshN1, 23 to 95 aa), which is followed by the cleavage site (PshN2, 23 to 148 aa) (Fig. 5A). We mutagenized this site into VIH/AAAAY (termed PshN2') and then performed the Y2H analysis. The results confirmed that CtsK1 (mainly the CtsK1C domain as shown above) lost its ability to target PshN2', confirming the targeting of the Psh cleavage site by CtsK1. It was also uncovered that CtsK1 does not interact with the C-terminal protease domain of Psh (Fig. 5B).

Based on the above results, we next reasoned and verified the assumption by yeast three-hybrid (Y3H) analyses that Fkp1 could

block the interaction between CtsK1C and Psh, especially between CtsK1C and PshN based on the results of yeast cell survival and β -galactosidase activity assays (Fig. 5C and D). To answer whether Fkp1 only binds CtsK1C or additionally attacks Psh to block its activation by CtsK1, we further performed the Y2H analysis. It was found that Fkp1 could positively interact with Psh through its PshN2 domain but not its PshN1 region nor its C-terminal protease domain. In addition, it was verified that Fkp1 could not interact with PshN2' (Fig. 5E), indicating its masking of the CtsK1-cleavage site on Psh. Protein pull-down analysis confirmed the interaction between Fkp1 and Psh (Fig. 5F).

Fungal subtilisin Pr1 is baited and sensed by Psh (9, 11). We finally performed the Y2H analysis and confirmed that the *Metarhizium* Pr1A could target Psh through its N terminus. The mutagenesis of the Psh cleavage site (i.e., PshN2') could not block its interaction with Pr1A (Fig. 5G), which verified that Pr1A targets the Psh motif (i.e., the bait region) being upstream of the CtsK1-cleavage site. We expressed both the Psh and Pr1A enzymes (SI Appendix, Fig. S3), and our biochemical experiments verified that the addition of different amounts of Fkp1 did not interfere with the cleavage of Psh by Pr1A (Fig. 5H). This result revealed that two fungal factors, Fkp1 and Pr1A, do not conflict with each other in their roles in insect infection.

Discussion

In this study, we report that *Metarhizium* fungi have evolved the effector Fkp1 to target the cathepsin protease CtsK1 and its cleavage-maturation site of Psh to block the activation of this antifungal protease cascade in *Drosophila*. This kind of dual targeting may result in the formation of three possible scenarios:

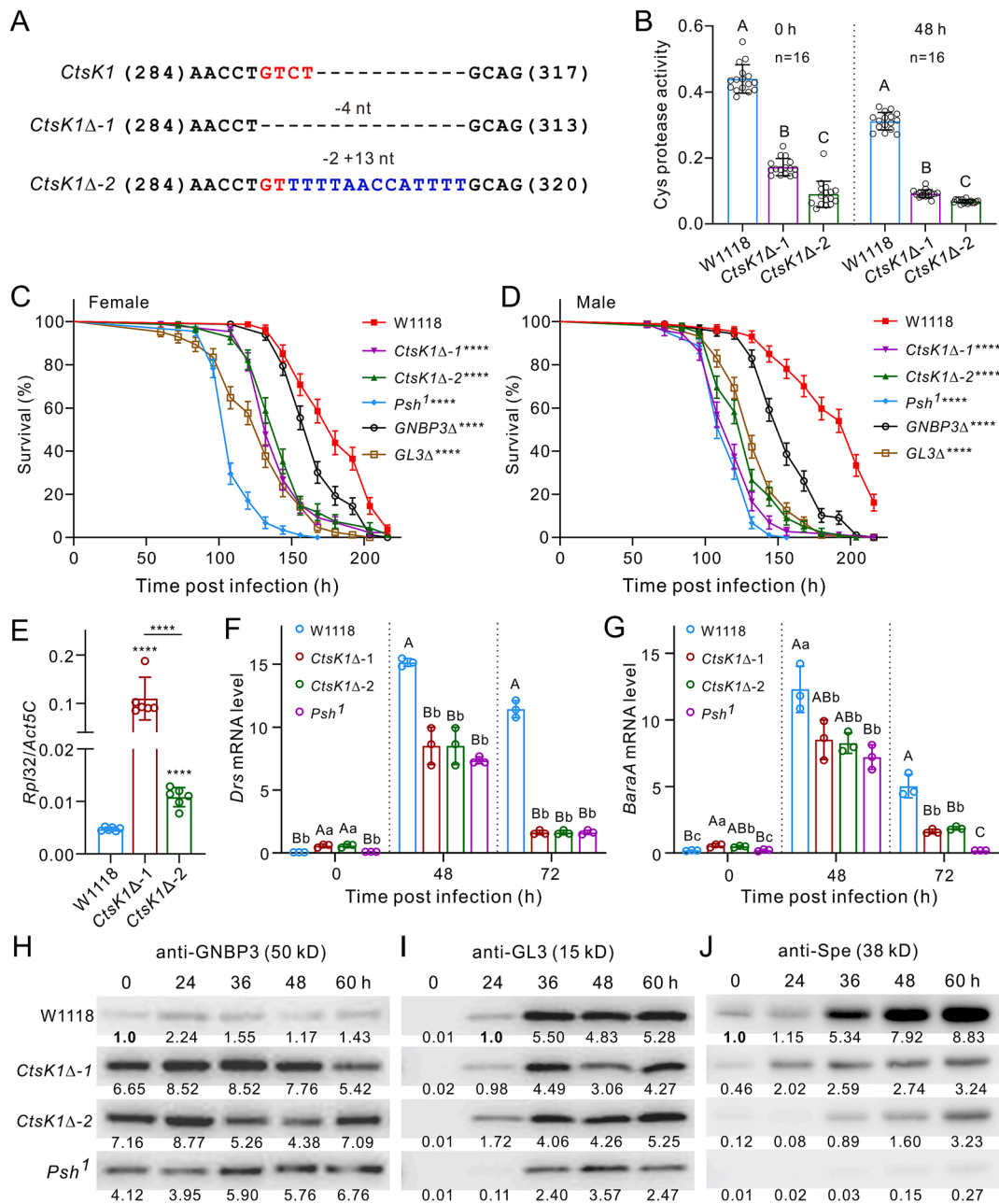


Fig. 3. *CtsK1* is essentially required in *Drosophila* for mediating antifungal immunity. (A) PCR sequencing verification shows the obtaining of two independent *CtsK1*-disruption fly lines. (B) *CtsK1* disruption substantially reduced the hemolymph CysP activity of *D. melanogaster* before and after fungal challenges. (C and D) Differential survivals of the female (C) and male (D) *D. melanogaster* lines after topical infection with the WT strain of *M. robertsii*. (E) Fungal load assays confirm the reduction of fungal colonization in *CtsK1Δ* mutant female flies after topical infection with the WT strain of *M. robertsii* for 84 h. Values are the mean $\pm$ SD. Pairwise Student's *t* test was conducted: *****p* < 0.0001. (F and G) *CtsK1* disruption significantly reduced the expression of antifungal genes *Drs* (F) and *BaraA* (G) in *Drosophila* females after topical challenge with *M. robertsii* for different durations. The log-rank test was conducted between the WT W1118 and individual mutant fly lines. (H–J) Differential accumulation of the immune factors GNBPs (H), GL3 (I), and Spe (J) in the wild-type and mutant female flies after topical fungal infections. The *Psh1* flies were included as a control. Protein intensities were quantified using ImageJ, and the values labeled below each band were calculated by reference to the band value labeled as 1.0 in bold in each panel.

CtsK1–Fkp1, Fkp1–Psh, and/or *CtsK1*–Fkp1–Psh complexes, all of which would suppress the maturation and activation of Psh (Fig. 5*I*). The *Fkp1*-like genes are alternatively present in EPF species including *B. bassiana*, and we have verified that BbFkp1 can also bind *CtsK1*. Taken together with the previous finding of targeting the β -glucan coreceptors GNBPs and GL3 by the *Metarhizium* Tge1 (8), our data show that insect pathogenic fungi have evolved divergent effectors to block both the PAMP receptors and sensing protease to counterattack the host dual recognition of fungal infections (Fig. 5*J*). The results of this study strongly reflect the arms-race coevolution between EPF and insects. It is

still enigmatic regarding the patchy distribution of *Fkp1*-like genes in alternative insect pathogens.

The antifungal Toll pathway was first established in *D. melanogaster* by injection and induction with the human pathogenic fungus *A. fumigatus* (30). It is true that, however, mammalian and plant pathogenic fungi can rarely infect or kill flies and other insects in nature. Even as animal pathogenic fungi, ascomycete EPF including *Metarhizium* and *Beauveria* species are evolutionarily closer to plant pathogens rather than mammalian pathogenic fungi (31, 32). In support to some extent, similar to the presence of the *Tge1*-like gene in phytopathogenic fungi but absent in

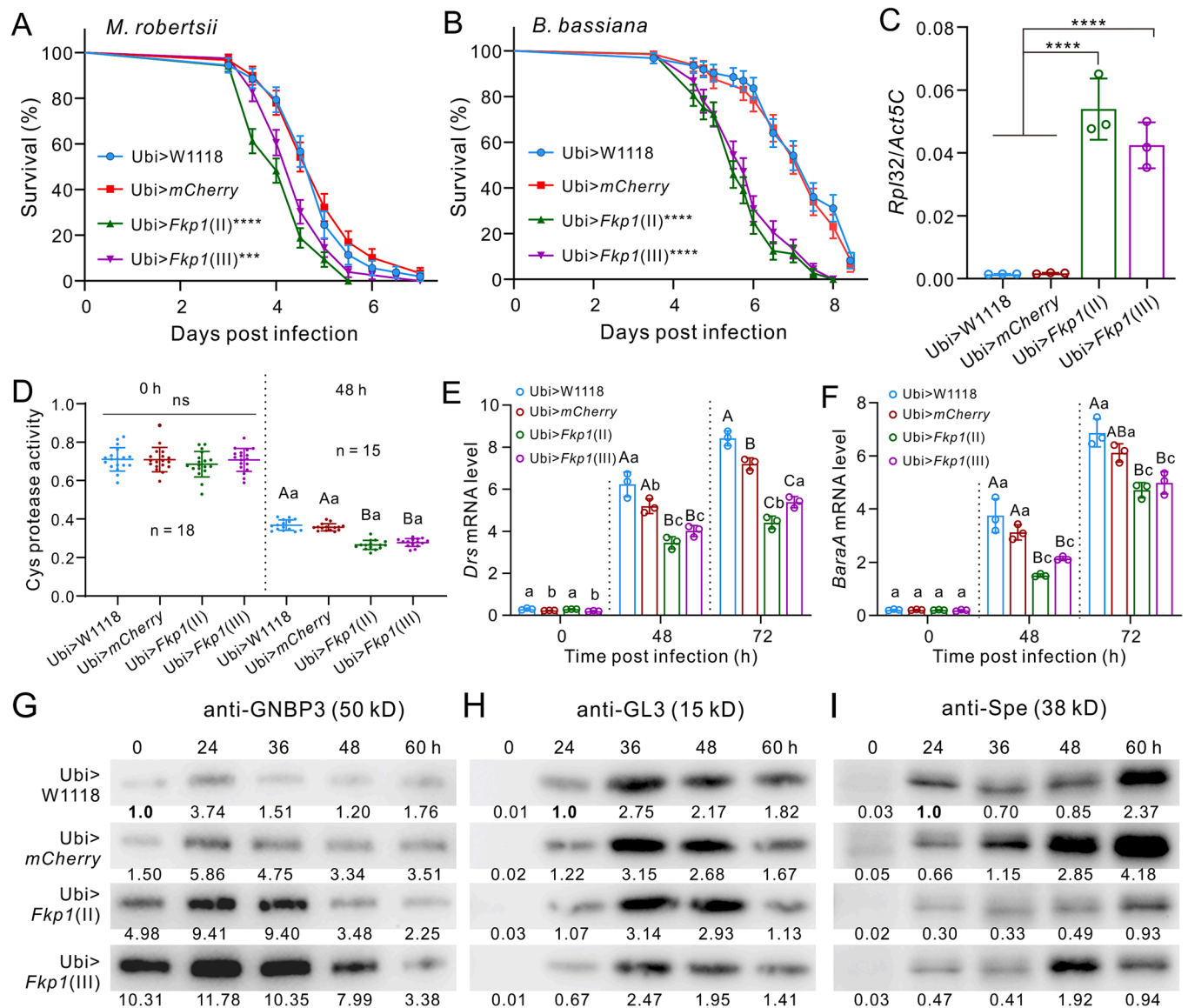


Fig. 4. Transgenic expression of *Fkp1* in *D. melanogaster* suppresses fly antifungal immune responses. (A and B) Transgenic insertion of *Fkp1* either in the chromosome II or III of *D. melanogaster* for expression reduced female fly survivals after topical challenge with the WT strains of *M. robertsii* (A) or *B. bassiana* (B). (C) Fungal load assays show that transgenic expression of *Fkp1* in *Drosophila* boosted fungal colonization of flies after topical fungal infection with *M. robertsii* for 84 h. Values are the mean $\pm$ SD. Pairwise Student's *t* test was conducted: *****P* < 0.0001. (D) Transgenic expression of *Fkp1* in *Drosophila* substantially reduced the hemolymph CysP activity after fungal infections. (E and F) *Fkp1* transgenic flies were impaired in expressing *Drs* (E) and *BaraA* (F) after topical fungal challenge with *M. robertsii* for different durations. (G–I) Differential accumulation of the immune factors GNBP3 (G), GL3 (H), and Spe (I) in the mock controls and *Fkp1*-transgenic female flies after topical fungal infections. Protein intensities were quantified using ImageJ by reference to the band value labeled as 1.0 in bold in each panel.

animal fungi (8), *Fkp1* homologues are also present in the closely associated plant pathogens but not in mammalian pathogens. The *Tge1*-like effector has also been verified with a virulence requirement in the rice blast fungus *M. oryzae* (33). Considering that the papain-like CysPs are widely present in plants, it is plausible, though yet to be determined, that *Fkp1*-like genes in phytopathogens may play a similar role in blocking the function of immune CysPs in plants.

As aforementioned, multiple CysP-inhibiting effectors have been identified in plant pathogens. We identified *Fkp1* as a blocking effector of *Drosophila* cathepsins. We unveiled that both *Fkp1* and *BbFkp1* can potentially target CtsB, CtsL1, CtsL3, Swim, CtsK2, and CtsF. Likewise, the *Ustilago* effector Pit2 can bind multiple CysPs in maize apoplasts (16). Our comparative enzyme activity assays of the WT and *CtsK1Δ* flies suggested that CtsK1 would be the major circulating cathepsin in *Drosophila*. Aside from the evidence that CtsB is not required in flies for mediating

antifungal defense, other cathepsin genes' immune contribution might be marginal. Functional investigations have shown that CtsL1 (previously called CP1) is required for autolysosomal clearance in the larval midgut cells of *Drosophila* (34). The cathepsin Swim can bind to the morphogen Wnt/Wingless (Wg) to maintain the solubility of Wg for promoting long-range signaling (35). CtsK2, being quite divergent from CtsK1, has been implicated in the regulation of sexual reproduction in *Drosophila* (36). Natural infection of insects by EPF like *Metarhizium* and *Beauveria* species can lead to insect death within a couple of days (37). It is possible, though unlikely, that *Fkp1* might also influence insect development by targeting these immune-irrelevant cathepsins.

Previous studies have shown that disrupting *GNBP3* leads to the upregulation of *Psh*, and vice versa, in *Drosophila* during fungal challenges (8, 9), which implies a compensatory pattern of immune recognition. In support to some extent, we found that *CtsK1* disruption or *Fkp1* transgenesis resulted in the upregulation

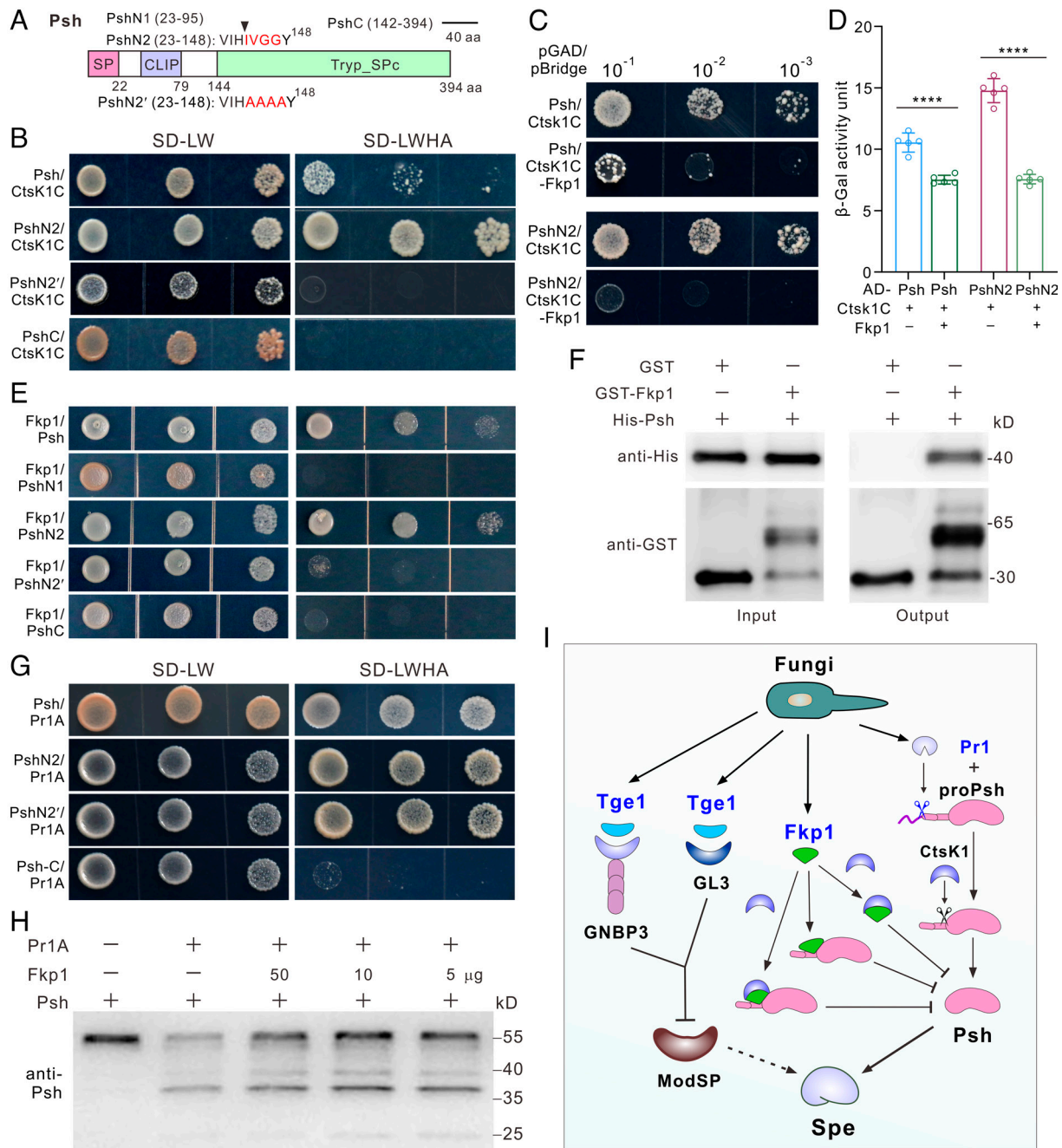


Fig. 5. Fkp1 blocks the catalytic activation of Psh by CtsK1. (A) Schematic structuring of the sensing protease Psh. Different regions or site mutations are indicated for the following experiments. (B) Y2H analysis shows the targeting of the Psh N-terminal cleavage site by CtsK1. (C and D) Y3H analysis confirms the Fkp1 inhibition of the Psh/CtsK1 interaction by cell survival (C) and β -galactosidase activity (D) assays. Pairwise Student's *t* test was conducted to compare enzyme activity between samples: *****P* < 0.0001. (E) Y2H analysis confirms the targeting of the CtsK1 cleavage site on Psh by Fkp1. (F) Protein pull-down analysis confirms the interaction between Fkp1 and Psh. (G) Y2H analysis shows that Pr1A targets the Psh N-terminal bait region without being influenced by CtsK1 cleavage site mutagenesis. (H) Enzymatic cleavage analysis shows that adding different amounts of Fkp1 has no obvious negative effect on Pr1A digestion of Psh. (I) Schematic pathway showing the inhibition of *Drosophila* antifungal immunity by fungal effectors. Fkp1 contributes to blocking the Psh maturation by targeting CtsK1 and Psh at its cleavage site in three scenarios.

of the surveillance GNB3 but not the inductive GL3 receptor. Providing that they play dual recognition roles, our survival data unveiled that the CtsK1–Psh protease cascade is more critical than GNB3 in sensing pathogen attacks and protecting flies against natural fungal infections. In support, it has been shown that *Drs* expression was not induced in *Psh*¹ (with a functional GNB3) but highly activated in the *GNB3*^{hades} (with a functional Psh) flies after topical infection by *B. bassiana* for 48 h (9). As shown in this study and supported by current knowledge, following the detection and preliminary cleavage by the fungal protease Pr1A, Psh is matured by CtsK1, which subsequently processes Spe for

maturation of the Toll ligand Spz (38). Instead, the stepwise-processing downstream factors of GNB3 identified so far include the modular protease ModSP and the clip-domain proteases cSP48 and Grass, which act upstream of Spe (10, 29, 39). In this context, the protease-sensing pathway may elicit a faster response than the GNB3-receptor cascade. Nevertheless, the fine-tuning machinery of these binary modules remains to be elucidated. The reduced accumulation of Spe in the *CtsK1*Δ and *Fkp1*-transgenic flies might reflect a feedback mechanism. Taken together with the evident downregulation of antifungal peptides, the subsequent processing of the Toll ligand Spz by Spe might be

impaired or disrupted in the *CtsK1* disruption and *Fkp1*-transgenic flies after fungal infections.

Having shown that Fkp1 targets the CtsK1 peptidase domain, it is not surprising that this fungal effector can attenuate the interaction between CtsK1 and Psh. Interestingly, our subsequent experiments uncovered that Fkp1 also binds Psh by targeting its CtsK1-cleavage site. This kind of dual targeting may more effectively ensure the inhibition of this protease cascade. In support, a small compound designed to target the MoErs1–OsRD21 interface has been shown to effectively combat rice blast disease (18). It remains to be investigated whether the Fkp1 dual binding processes occur sequentially or simultaneously. Future efforts are also required to determine the complex structure, whether in dimers or as a CtsK1–Fkp1–Psh trimer.

In summary, we report the identification of the Fkp1 effector from *M. robertsii* that targets the *Drosophila* danger-sensing protease cascade to suppress fly antifungal immune defenses. Taken together with the previous discovery of the PAMP-blocking factor (8), our results reveal that insect pathogenic fungi have evolved distinct effectors to block host immune recognitions, which paints an interesting picture of the effector mechanisms in mediating fungus–insect interactions.

Materials and Methods

Microbial Strains and Maintenance. The WT strains ARSEF 2575 of *M. robertsii* (40) and ARSEF 2860 of *B. bassiana* (41) were used in this study. Fungal strains were cultured on potato dextrose agar (PDA; BD Difco) at 25 °C. The WT strain of *M. robertsii* was also grown in Sabouraud dextrose broth (SDB; BD Difco) at 25 °C and 200 rpm for three days for RNA extraction. The WT conidia were inoculated on the black soldier fly (*Hermetia illucens*) wings placed on a water agar for 30 h for appressorium formation, and the hyphal body cells were isolated from the hemolymph of wax moth (*G. mellonella*) larvae by gradient centrifugation in Percoll (Sigma-Aldrich) after injection with fungal conidia for 36 h (42). These samples were also used for RNA isolations. Different strains of *Escherichia coli* were used for gene cloning (Top10, Weidi Biotech) or protein expressions (Rosetta) by maintaining either on the Luria-Bertani (LB) agar at 37 °C or in LB broth at 16 °C. The budding yeast *Saccharomyces cerevisiae* strains Y2H Gold, AH109, and YTK12 (Weidi Biotech) were maintained on the synthetic dropout (SD) medium (Sigma-Aldrich) and used for Y2H, Y3H or protein secretion analysis (8). The methylotrophic yeast *Pichia pastoris* (now *Komagataella pastoris*) strain GS115 (Thermo Fisher Scientific) was maintained on the nutrient-rich YPD (1% yeast extract, 2% peptone, and 2% dextrose) agarose plates, and used for expression of proteases.

***Drosophila* Stocks and Maintaining.** Different lines of *D. melanogaster* were used in this study, including the wild-type W1118, and mutant lines *Psh*¹ (27), *CtsB* (Bloomington Stock BDSC# 15434), *GNBP3Δ* and *GL3Δ* (8) as well as the *CtsK1* mutants generated in this study. Different tool lines mentioned below were used to generate isogenic null mutants or transgenic flies. The flies were maintained at 25 °C and 12 h of light/dark cycles on the Bloomington formulation of cornmeal agar medium (43).

Fungal Gene Deletion and Complementation. Our previous has shown that deletion of the effector-like gene *EXU99903* (now *Fkp1*) in *M. robertsii* impaired fungal virulence against insects (8). In this study, the *Fkp1* homologous gene in *B. bassiana* (*BbFkp1*, *BBA_10113*) was deleted by homologous replacement. The 5′- and 3′-flanking regions were amplified using different primer pairs (SI Appendix, Table S4) and the 2× Phanta Master Mix PCR Kit (Vazyme). The purified products were cloned into the binary vector pDHT-SK-Bar for fungal transformation (44). The null mutants of both fungi were complemented by amplifying the full open reading frame of the target genes together with each promoter region (1.5 kb upstream of the start codon). The purified product was cloned into the plasmid pDHT-SK-Sur (conferring resistance to sulfonyleurea) (45), and the vectors were subjected to transform the null mutant of each fungus. *Fkp1* was also overexpressed in the WT strain of *M. robertsii* under the control of the *Tef* promoter (46).

The drug-resistant mutants were selected for single spore isolations and verified by PCR and/or RT-PCR (SI Appendix, Table S4).

Characterization of Fungal Growth and Infection Features. Both the WT and mutant strains of *M. robertsii* were grown on PDA for two weeks to determine whether there were any growth defects after the deletion of *Fkp1*. The differentiation of infection structure was induced on Petri dishes (6 cm in diameter) containing 3 ml of 0.0125% (g/v) yeast extract. After inoculation of 10 μl spore suspension (2×10^7 conidia/ml in 0.05% Tween 20), the plates were incubated at 25 °C for 20 h (47). Fungal penetration assays were conducted by inoculating the center of cellophane membrane (Dingguo Biotech) with a droplet of spore suspension (2 μl of 2×10^6 conidia/ml). The membranes were carefully removed from the minimum medium agar five days postinoculation, and the plates were kept for incubation at 25 °C for four days (42).

Both the WT and mutant strains of *M. robertsii* and *B. bassiana* were subjected to topical infection of the females and/or males of *D. melanogaster* and *D. suzukii* by immersion of anesthetized insects in spore suspension (2×10^5 conidia/ml for *M. robertsii* and 2.5×10^7 conidia/ml for *B. bassiana*) for 30 s (8). The insects treated with 0.05% Tween 20 were used as a mock control. The mutant flies generated as described below were also subjected to survival assays using the WT strains of either *M. robertsii* or *B. bassiana*. More than 70 insects were used for each treatment, and the experiments were repeated at least twice. The mortality of insects was recorded every 12 h. The difference in insect survival was determined by the log-rank test using GraphPad Prism (ver. 10.1.1). The results from the same batch of experiments were shown at the end. Fungal load assays were conducted by counting fungal CFUs and/or qPCR analysis of fungal DNA copies in flies as described before (26).

Screening and Verification of the Fkp1 Targeted Proteins in *D. melanogaster*. We used the previously generated immune-related cDNA library of *D. melanogaster* for screening potential proteins targeted by Fkp1 (26). The cDNA of *Fkp1* was cloned into the yeast plasmid pGBKT7 (TaKaRa Bio) without its signal peptide sequence. This bait plasmid was subsequently transformed into the yeast strain Y2H Gold for screening the *Drosophila* cDNAs harbored in the pGADT7 vectors (26). Positive clones were obtained from the SD medium (SD-Trp-Leu-His-Ala; TaKaRa) plus aureobasidin A (AbA; Yeasen) with or without X-Gal (Beyotime). Yeast plasmids were then extracted from these positive clones for sequencing using the universal T7 primer. The obtained sequences were subjected to BLAST analysis.

The sequences without frameshift from the above positive clones belong to different genes (SI Appendix, Table S1), which were then subjected to further verification by cloning their full cDNA sequences for Y2H analysis (8). The homologous *BbFkp1* of *B. bassiana* was also used for Y2H verification. CtsK1 and CtsB proteins contain a cathepsin inhibitor domain I29 and a C-terminal peptidase C1A domain. Both Fkp1 and BbFkp1 were then subjected to interactive analysis with each protein's N- (the sequence following the SP to I29 region) and C termini (only the C1A domain). Additional cathepsin genes *CtsL1* (CG6692), *CtsL3* (CG11459), *CtsK2* (CG4847), *Swim* (CG3074), and *CtsF* (CG12163) of *D. melanogaster* were also cloned and included in Y2H analysis using Fkp1 and BbFkp1 as individual baits. Considering that CtsK1 mediates the maturation of Psh (11), Fkp1 was also subjected to Y2H analysis with the full-length, N- and C-terminal regions of Psh. In particular, the cleavage site (141-VIHVGGY-148, PshN2) of Psh targeted by CtsK1 was mutagenized (141-VIHIAAAAY-148, PshN2′) by gene synthesis (Beyotime), and used for interactive analysis.

Phylogeny Analysis. Besides Fkp1 and BbFkp1, our BLASTp analysis indicated that the orthologs of *Fkp1* are also present in other EPF species and ascomycete plant and nematode pathogenic fungi based on the cutoff E-value $\leq 1e-20$ and amino acid identity $>40\%$. The homologous protein sequences were then retrieved for phylogenetic analysis. There are ten proteins termed cathepsins cataloged in FlyBase (SI Appendix, Table S2). The amino acid sequence of each protein was then retrieved, and the obtained protein sequences were aligned with the program Clustal X (48). The phylogenetic relationship of these proteins was inferred by using the maximum likelihood method and JTT matrix-based model. The analysis was conducted with 500 bootstrap replications in MEGA11 (49).

Protein Expression, Pull-Down and Degradation Analysis. Different proteins were heterologously expressed in different cells for verification of protein interactions and/or proteolytic analysis. Thus, *Fkp1* was fused in-frame with the glutathione-S-transferase (GST) tag without its signal peptide region, and the cassette was cloned into the vector pGex-6p-1 (Addgene). The C-terminal C1A domain of *CtsK1* was cloned in fusion with a His (6×) tag into the pET-28b vector. The plasmids were individually transformed into the *E. coli* Rosetta strain for expressions and purifications as we described before (8). In addition, *Psh* was cloned into the pFastBac1 vector (Addgene) in fusion with a His tag. The bacmid was amplified in the *E. coli* DH10 Bac strain, purified, and used to transfect the Sf9 insect cells, which were incubated in the Sf-900 II SFM medium (Thermo Fisher) for three days at 28 °C and 120 rpm. The cell suspension was centrifuged at 4 °C and 8,000 rpm for 10 min, and the supernatant was used for protein purification. Immunoblotting analysis was performed using the mouse monoclonal antibodies (Abs) against the GST and His tags (Proteintech), respectively. The antigenic peptide (NISYAEQPGSIRLLK, 301 to 115 aa) of *Psh* was predicted, synthesized, and used to raise the polyclonal antibodies in rabbits (Youke Biotech, Shanghai).

The Pr1A protein (MAA_05675) of *M. robertsii* was expressed using the *Pichia pastoris* strain GS115 (Thermo Fisher Scientific) by following the manufacturer's protocol. The *Pr1A* cDNA was amplified in fusion with a C-terminal His tag, and the purified product was cloned into the plasmid pPIC9K for yeast transformation and protein expression. The purified Pr1A was used to raise polyclonal antibodies in rabbits (Youke Biotech, Shanghai) and for proteolytic treatment of pro-*Psh* in the presence or absence of *Fkp1* in a Tris buffer (50 mM, pH 7.9). The reaction system (15 µl) includes pro-*Psh* (30 µg), Pr1A (50 µg), and *Fkp1* (at a final dosage of 0, 5, 10, and 50 µg, respectively), and the reaction was kept at 28 °C for 30 min. The samples were then boiled in SDS-loading buffer (5 µl) for 5 min before use for protein gel separation and Western blot (WB) analysis using the anti-*Psh* peptide antibody raised above.

Generation of Mutant Flies. The *Ctsk1* null mutant flies of *D. melanogaster* were generated at the National Drosophila Resource Center of China (<http://ndrcc.sibcb.ac.cn/ndrcc/>) using the CRISPR-Cas9 technique (50). Two single-guide RNAs *CtsK1*Sg1 and *CtsK1*Sg2 (*SI Appendix, Table S4*) were synthesized and used for injection of the embryos of the Bloomington Stock fly BDSC #78781 [$y^1 sc^+ v^1 sev^{21}$; P(nos-Cas9.R)attP40]. Successful deletion of *CtsK1* was confirmed by PCR sequencing. The flies were crossed to the W1118 background, and the isogenic male and female flies were further self-crossed for more than five generations before being used for experiments (8). To generate the *Fkp1* transgenic flies, the cDNA of this gene was cloned into the plasmid pUAST-attB, and the vector was used to inject the embryos of both $y^1 w^{67} c^{23}$; P(CaryP)attP40 and $y^1 w^{67} c^{23}$; P(CaryP)attP2 (BDSC# 8622) flies for integration into either the Chr II or Chr III of *D. melanogaster*. The obtained isogenic male flies were genetically crossed with the virgins of the *Ubi-gal4* (*w*; *Ubi-Gal4*; *Tub-Gal80^{ts}*) or *Da-gal4* (*w*; *Da-Gal4*; *Tub-Gal80^{ts}*) flies. The F1 progeny of female or male flies was maintained at 29 °C for three days before use for experiments. The *UAS-mCherry* and W1118 male flies were similarly crossed with the driver virgins for use as reference controls. After topical infection of these flies with the WT strain of *M. robertsii* (2×10^5 conidia/ml) for different durations, fly hemolymph samples were collected (25 males and 25 females each) using the Zymo-Spin IC column (Zymo Research) for enzyme activity assays (51), and WB analysis of GNBP3, GL3, and *Spe* expressions. The GNBP3, GL3, and *Spe* antibodies were generated before (8). The intensity of each protein band was measured using ImageJ (ver. 1.53) (52).

Quantitative Analysis of Gene Expressions. To determine the expression of *Fkp1*, fungal samples prepared above were used for RNA extraction using the TransZol Up RNA Kit (Yeasen Biotech). After the fungal challenge of different

flies indicated above, the expression of the antifungal genes *Drs* and *BaraA* was determined. We also examined the expression of the β -glucan receptor genes *GNBP3* and *GL3* in *Drosophila* after deletion of *CtsK1*. RNA extraction, cDNA conversion, and quantitative PCR analysis were performed as described before (8). There were three independent replicates for each sample, and the experiments were repeated at least twice. The data from the same batch of experiments were used as the final representatives for one-way ANOVA comparative analysis with Tukey's test.

Cysteine Protease Activity Assay. Cysteine protease activity assays were conducted for different hemolymph samples by using the substrate Z-Phe-Arg-AMC (7-amino-4-methyl coumarin; at a final concentration of 50 µM; Bachem), a broad-spectrum substrate of cathepsins (22). Protein concentrations were determined using a BCA Protein Assay kit (Thermo Fisher Scientific). The protease inhibitor E-64 (at a final concentration of 10 µM; Sigma-Aldrich) was used for pre-incubation with hemolymph as a reference control. The hemolymph samples were collected from different flies (80 each: 40 males and 40 females) at 4 °C using the spin columns after topical fungal challenge for different times (0, 36, 48, 60, and 72 h). Each sample (5 µl) was incubated in 192 µl of activation buffer (100 mM sodium phosphate buffer; 1 mM EDTA; 4 mM dithiothreitol; pH 6.0) for 10 min before the addition of 2 µl of Z-Phe-Arg-AMC in a 96-well microtiter plate (Corning). The plate was incubated under gentle agitations at 28 °C for 30 min before measuring the fluorescence release of AMC using a microplate reader (Thermo Varioskan Flash) at the wavelength of $\lambda_{exc} = 350$ nm and $\lambda_{em} = 460$ nm. The unit activity of cysteine protease was determined as the AMC fluorescence unit released per minute per microgram of protein (22). There were 16 or 18 parallel replicates for each sample, and the difference between samples was determined by one-way ANOVA coupled with Tukey's test.

Yeast Three-Hybrid Analysis. To determine whether *Fkp1* could block the interaction between *CtsK1* and *Psh* N terminus, we performed the Y3H analysis based on the yeast handbook PT3024-1 (Clontech). In brief, the *Psh* and *PshN2* cDNAs were cloned into the pGADT7, and *CtsK1C* was cloned into the multiple cloning site (MCS) I of pBridge (Clontech, TaKaRa Bio). *Fkp1* was then cloned into the MCS II site of pBridge under the control of the Met-repressible *pMET25* promoter. All plasmids were confirmed by PCR sequencing and transformed into the yeast AH109 strain. The transformed yeast cells were incubated on the SD-Leu-Trp-Met medium (TaKaRa Bio) at 30 °C for 4 d. Individual colonies were then spotted on the SD-Trp-Leu-Met, SD-Trp-Leu-His-Met (plus 1 mM 3-AT) and SD-Leu-Trp-His (plus 1 mM 3-AT) media (TaKaRa), respectively. The inhibition of *Fkp1* on *CtsK1C* and *Psh*/*PshN2* interactions was determined by yeast survival assays or using the substrate chlorophenol red- β -D-galactoside (CPRG, Sigma-Aldrich) to measure β -galactosidase activity.

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

ACKNOWLEDGMENTS. We appreciate Dr. Wei Wu's help in generating fly mutants. This work was supported by the National Natural Science Foundation of China (Nos. 32230087 and 32021001) and the National Key R&D Program of China (2022YFD1400700 and 2022YFD1400500).

Author affiliations: ^aKey Laboratory of Insect Developmental and Evolutionary Biology, CAS Center for Excellence in Molecular Plant Sciences, Shanghai Institute of Plant Physiology and Ecology, Chinese Academy of Sciences, Shanghai 200032, China; ^bInstitute of Edible Fungi, Shanghai Academy of Agricultural Sciences, Shanghai 201106, China; ^cSchool of Life Science and Technology, ShanghaiTech University, Shanghai 201210, China; and ^dCAS Center for Excellence in Biotic Interactions, University of Chinese Academy of Sciences, Beijing 100049, China

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